Federated Intrusion Detection Under Label-Flipping Poisoning

A simulated ten-client study on CICIoT2023

Salvador Leon Lopez · Elena Victoria Sanchez · Joe Reyna · April Flores

MILESTONE 1 OF 4: LOADING RESULTS...



What does one poisoned client cost?

research_question.txt

In a ten-client federated IDS trained on non-IID partitions of CICIoT2023, how much detection performance is lost when one client flips its labels, and does **trimmed-mean aggregation** recover it?

H1 · under FedAvg

> 3 POINTS

drop in global macro F1 under FedAvg, versus the paired clean run

H2 · under trimmed mean

>= 50% BACK

of that loss recovered. Smaller effects falsify the hypothesis.

TAKEAWAY > If FedAvg loses over 3 F1 points and trimmed mean wins back half, the hypothesis holds

C:\GROUP8> 02_how_it_works.exe

One federated round, repeated 30x

1 · SEND



Server sends the global model to all 10 clients

2 · TRAIN



Each client trains one epoch on its own private data

3 · RETURN



Clients send back model weights, never raw traffic

4 · COMBINE



Server merges them: FedAvg or trimmed mean

`> honest_client.labels`

DDoS | Recon | Mirai | Benign

`> attacker.labels (half its attacks flipped)`

DDoS → Benign | Recon | Mirai → Benign | Benign

TAKEAWAY > The server only sees weights, so it cannot see that the attacker’s labels are wrong

C:\GROUP8› 03_dataset.csv

CICIoT2023, cleaned and subset

105
IoT devices, one lab testbed

39
numeric flow features

33 → 7
attack types into the authors' families, plus benign

23.6K
flows in our pilot (12 of 63 files, ≤ 3,000 per family)

› pipeline.bat · runs in this order

01 READ
All CSVs; label counts before sampling

→

02 CLEAN
Drop inf/NaN, duplicates, label conflicts

→

03 CAP
Cap per family: 3,000 pilot, 45,000 full

→

04 SPLIT
Train / val / test, grouped by source file

→

05 FIT
Scaler and partitions on train only

TAKEAWAY › We clean before splitting, so no duplicate can leak between training and test data

C:\GROUP8> compare_datasets.exe

Why CICIoT2023?

RECENT | IoT-SPECIFIC | MANY ATTACK FAMILIES | PUBLIC AND LABELED

Dataset	Setting	Attacks	For our study
CICIoT2023	105 IoT devices, 2023	33 attacks, 7 families	Chosen
CIC-IDS2017	Enterprise network, not IoT	About 14 attack labels	Older; known labeling errors
TON_IoT	IoT and IIoT telemetry	9 attack types	Fewer families, mixed sources
N-BaIoT	9 IoT devices, 2018	2 botnets only	Too few families to flip

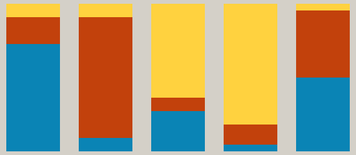
TAKEAWAY › Enough attack families to make flipping meaningful, on modern IoT traffic

C:\GROUP8› 04_partitions.exe

Ten simulated clients, three ways

PRIMARY

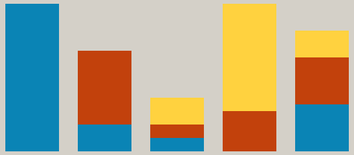
Dirichlet, α = 0.5



Each client gets a different family mix: controlled skew

ALTERNATIVE

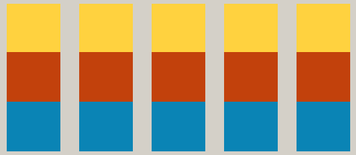
k-means, k = 10



Label-free clusters; risk of unequal or single-family clients

CONTROL

IID



Every client gets the same mix and size

› checks.log (every client)

[OK] ≥ 2% of the training pool

[OK] ≤ 95% from one family

[OK] ≥ 2 learnable families

partition_checks.log · pilot data

Split	Rows per client	Max purity	Result
Dirichlet	787 – 2,778	50%	PASS
k-means	1 – 4,804	100%	FAIL
IID	1,753 – 1,754	15%	PASS

TAKEAWAY › k-means failed our checks (a 1-row client, 100%-pure clients); Dirichlet passed

C:\GROUP8› 05_tools.ini

A small, inspectable stack

MS-DOS Prompt

```
C:\› stack
Python · PyTorch · scikit-learn · pandas (pinned)
C:\› fl_framework
Our own FL loop, not Flower: the spec allows it,
Flower 1.39 deprecated its simulation API, and every
step stays visible
C:\› pytest
23 passed · aggregation, attack, end-to-end
```

SAME MODEL IN EVERY CONDITION

MLP 39 → 128 → 64 → 8

Setting	Value
Activation, dropout	ReLU, 0.1
Optimizer	Adam, lr 0.001
Batch size	256
Federated	30 rounds × 1 epoch, all 10
Traffic per round	1.11 MB

TAKEAWAY › One fixed model and our own loop: only partition, attack and aggregation change

C:\GROUP8› 06_attack_and_defense.exe

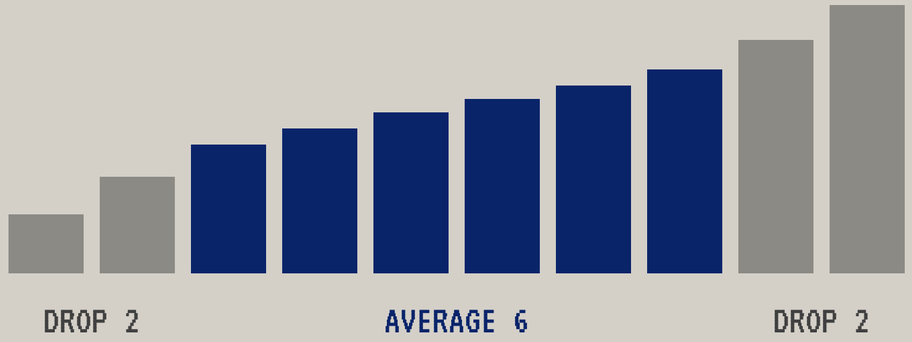
A fixed, reproducible threat model

ATTACK

- Attacker: client with most attack rows
- Flips 50% of each family to benign
- Same poisoned rows every round
- Changes labels only, trains honestly
- Sweep: 1, 2, 3 attackers

DEFENSE

β = 0.2 · per weight, sort the 10 client values:



TAKEAWAY › The attack is fixed and repeatable; the open question is whether trimming catches it

C:\GROUP8› 07_changelog.txt

What changed since the proposal

v1.0 6 clients · k-means · Flower → v2.0 10 clients · Dirichlet · own loop

Feedback	Our change
K-means may give unequal, label-pure clients	Dirichlet split is primary; every client checked
Use 10 to 20 clients; vary malicious share	10 clients; 1, 2 and 3 attackers
Deduplicate and group before splitting	Duplicates removed first; split grouped by file
Make the attack deterministic	Fixed attacker; poisoned rows fixed per seed
Specify the trimmed mean	β = 0.2: drop 2 per side, average 6
Pair seeds; report per client and class	Same initial weights per seed; support on every rate
Start with a small pilot	Pilot config: 12 files, 5 rounds, 1 seed

TAKEAWAY › Every feedback point is now implemented in code, not just promised

C:\GROUP8› 08_status.exe

Pilot runs on real data

Copying project... milestone 1 of 4

OCT 25 · DONE

Subset, partitions, FL loop, attack, defenses

NOV 15

Clean vs. poisoned runs, attacker sweep

NOV 30

Final presentation and demo

DEC 7

Final report; test set used once

› pilot_results.log · val macro F1

Centralized 0.647

FedAvg IID 0.526

FedAvg non-IID 0.434

FedAvg + attacker 0.312

Trimmed + attacker 0.374

› limits.txt

One lab testbed; simulated clients

Capped subset, not natural prevalence

Pilot: 12/63 files, 5 rounds, 1 seed

TAKEAWAY › Pilot: the attack cost 12.2 F1 points; trimmed mean won back half

C:\GROUP8› 09_lessons.txt

What the pilot taught us

upgrade.exe · pilot → full run

Setting	Pilot	Full run
Rounds	5	30 (curves still rising)
Files	12 of 63	All 63
Rows per family	3,000	Up to 45,000
Seeds	1	3, paired

Short, small and single-seed: every pilot number is an underestimate with no error bar

› to_investigate.txt

[?] Clean: trimmed beat FedAvg (0.47 vs 0.43)

[!] Clean: trimmed missed more (FNR 0.24 vs 0.09)

[!] Attacker = largest client: worst case

[+] Add per-family errors, median, 2–3 attackers

TAKEAWAY › Same design, bigger runs: we change scale and seeds, not the method

C:\GROUP8> exit

QUESTIONS?

Salvador Leon Lopez · Elena Victoria Sanchez · Joe Reyna · April Flores